



Published in final edited form as:

Clin Radiol. 2024 April ; 79(4): e607–e615. doi:10.1016/j.crad.2023.12.022.

Early change in apparent diffusion coefficient as a predictor of response to neoadjuvant androgen deprivation and external beam radiation therapy for intermediate- to high-risk prostate cancer

F.B. Franco^{a,†}, J.E. Leeman^b, A. Fedorov^a, M. Vangel^c, F.M. Fennessy^{a,*}

^aDepartment of Radiology, Harvard Medical School, Brigham and Women's Hospital, 75 Francis Street, Boston, MA 02115, USA

^bDepartment of Radiation Oncology, Harvard Medical School, Brigham and Women's Hospital, 75 Francis Street, Boston, MA 02115, USA

^cStatistician, General Clinical Research Center, Massachusetts Institute of Technology and Massachusetts General Hospital, 55 Fruit St, Boston, MA 02214, USA

Abstract

AIM: To determine the role of serial apparent diffusion coefficient (ADC) as a biomarker for response to neoadjuvant androgen deprivation therapy (nADT) followed by external beam radiation therapy (EBRT) in intermediate- to high-risk prostate cancer (PCa) patients.

METHODS: This Health Insurance Portability and Accountability Act (HIPAA)-compliant, institutional review board (IRB)-approved prospective study included 12 patients with intermediate- to high-risk PCa patients prior to nADT and EBRT, who underwent serial serum prostate-specific antigen (PSA) and multiparametric prostate magnetic resonance imaging (mpMRI) at baseline (BL), 8-weeks after nADT initiation (time point [TP]1), 6-weeks into EBRT delivery (TP2), and 6-months after nADT initiation (TP3). Tumour volume (tVOL) and tumour and normal tissue ADC (tADC and nADC) were determined at all TPs. tADC and nADC dynamics were correlated with post-treatment PSA using Pearson's correlation coefficient. Paired *t*-tests compared pre/post-treatment ADC.

RESULTS: There was a sequential decrease in PSA at all TPs, reaching their lowest values at TP3 post-treatment completion. Mean tADC increased significantly from baseline to TP1 ($917.8 \pm 107.7 \times 10^{-6}$ versus $1033.8 \pm 139.3 \times 10^{-6} \text{ mm}^2/\text{s}$; $p < 0.01$), with no subsequent change at TP2 or TP3. Both percentage and absolute change in tADC from BL to TP1 correlated with post-treatment PSA ($r = -0.666$, $r = -0.674$; $p = 0.02$). Post-treatment PSA in good responders (< 0.1

*Guarantor and correspondent: F. M. Fennessy, Department of Radiology, Harvard Medical School, Brigham and Women's Hospital, 75 Francis Street, Boston, MA 02115, USA. ffennessy@bwh.harvard.edu (F.M. Fennessy).

[†]Present Address: Radiology Associates of South Florida, Florida International University, 8900 N Kendall Dr, Miami, FL 33176, USA.

Conflict of interest

The authors declare no conflict of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.crad.2023.12.022>.

ng/ ml) versus poor responders (≥ 0.1 ng/ml) was associated with a greater increase in tADC from BL to TP1 ($169.2 \pm 122.4 \times 10^{-6}$ versus $22.9 \pm 75.5 \times 10^{-6}$ mm²/s, $p=0.03$).

CONCLUSION: This pilot study demonstrates the potential for early ADC metrics as a biomarker of response to nADT and EBRT in intermediate to high-risk PCA.

Introduction

Prostate cancer (PCa) is the second most common cancer in males and the fifth cause of death in men worldwide.¹ In the USA alone, approximately 191,930 new cases and 33,330 deaths were expected in 2020.² Patients presenting with high-risk features such as a high prostate-specific antigen (PSA), high Gleason score, and advanced local disease (T-stage T2b or higher) have a higher cancer mortality rate.^{3,4} For these patients, a combination of external beam radiotherapy (EBRT) with neoadjuvant, concurrent and adjuvant androgen deprivation therapy (nADT) has proven to be an effective method of improving survival as demonstrated by both the European Organization for Research and Treatment of Cancer (EORTC)⁵ and the Radiation Therapy Oncology Group (RTOG),⁶ amongst others.⁷⁻⁹ PSA has been widely accepted as a critical biomarker of response to nADT,^{10,11} with several studies showing that a PSA lower than 0.1 ng/ml during and after combined therapy is strongly associated with cancer-specific survival, biochemical progression-free survival, and overall survival¹²⁻¹⁴; however, there are known limitations to using early PSA as a biomarker of response, as in addition to production from PCa cells, there are known non-oncological contributions to PSA including normal prostatic production of PSA, benign prostatic hyperplasia, and significant physiological variability.¹⁵ Furthermore, patients who achieve an early PSA response are still at risk of treatment failure and progression. Additionally, a subset of de-differentiated high-grade PCa does not produce high levels of PSA making it an ineffective biomarker in some cases. There is no consensus on a definitive PSA value to stratify those in whom 6 months of ADT is sufficient for cure from those in whom longer androgen suppression should be pursued.¹⁶

Current interest is shown in multiparametric magnetic resonance imaging (mpMRI) of the prostate as a non-invasive biomarker for PCa aggressiveness¹⁷⁻²¹ and treatment response,²²⁻²⁴ utilising functional sequences such as diffusion-weighted image (DWI), and more specifically, DWI-derived apparent diffusion coefficient (ADC), which can provide information about tumour functionality and its microenvironment.²⁵⁻³⁰ Recently, studies have investigated the use of mpMRI as an imaging biomarker for neoadjuvant therapy response,^{24,31-35} some retrospectively showing a correlation between tumour ADC changes and PSA following nADT^{35,36} in a non-serial fashion. In a prospective study, Pasquier *et al.* correlated early ADC changes with PSA decrease during EBRT.³⁷ Nevertheless, although ADC change holds promise as a biomarker for clinical outcomes of nADT, the value of dynamic changes in ADC early in the course of combined nADT and EBRT to predict those who would fail combined therapy has not been explored. If early ADC metrics could successfully predict ultimate clinical outcomes, it may provide an extremely valuable biomarker for clinical decision-making, personalising radiation dose, length of ADT, and use of novel antiandrogen agents.

Through serial ADC measurements, the goal of this prospective pilot study was to investigate the use of dynamic changes in ADC as an early biomarker of response to combined nADT and EBRT and assess how it differs among the subpopulation of good clinical responders compared to poor responders to combined neoadjuvant therapy.

Materials and methods

This was a prospective single-institution study approved by the Institutional Research Board and was registered in [ClinicalTrials.gov](https://www.clinicaltrials.gov). Consecutive patients deemed suitable for neoadjuvant ADT and EBRT were consented prospectively between 2014 and 2015, and followed clinically for up to 5 years.

Patient cohort

Adult males were deemed eligible for the study if they had biopsy-confirmed intermediate- to high-risk localised PCa, identified as one or more of the following three categories: clinical or radiographic Stage T2b–T4 primary tumour; Gleason score 7–10 in any core on prostate biopsy, PSA >10 ng/ml before initiation of therapy. Patients were required to have visible disease on baseline MRI, were deemed suitable for combination therapy consisting of neoadjuvant androgen-deprivation therapy, and EBRT therapy per their treating oncologist and radiation oncologist and were able to provide informed written consent prior to study entry. Patients with prior localised or systemic treatment for PCa or those unable to undergo MRI imaging were excluded from the study. This study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki.

All enrolled patients underwent mpMRI at four different time points (TP), as outlined in Fig 1: at baseline (BL) before commencing treatment, after 8 weeks of nADT initiation (TP1), 6 weeks into EBRT delivery (TP2), and after 6 months of nADT initiation (TP3). Corresponding serum PSA was measured at all TPs.

Treatment plan

Androgen-suppression therapy consisted of bicalutamide and an LHRH-agonist (leuporelin; $n=10$), abiraterone acetate and leuporelin ($n=1$), and leuporelin ($n=1$) for 2 months prior to radiotherapy and continued for a minimum of 6 months depending upon the disease characteristics (i.e., T-stage, PSA, Gleason score), as decided by the treating oncologist, based on individual-risk categorisation. Image-guided intensity-modulated radiotherapy using implanted fiducials was planned to target the prostate gland, seminal vesicles, and in some cases, the pelvic nodes as well. All participants were treated with conventionally fractionated intensity-modulated radiation therapy to the prostate and seminal vesicles to a dose of 7,560 to 7,920 cGy in 42 to 44 fractions over 8–9 weeks.

Serum PSA

Baseline serum PSA was collected prior to initiation of therapy. For TP1–3, serum PSA was obtained on the same day as the corresponding MRI. PSA density was calculated based upon three-dimensional (3D) measurements on MRI and the corresponding PSA at that TP. Dynamic ADC changes were correlated with PSA at the end of therapy and how it differed

among the subpopulation of good clinical responders (PSA <0.1 ng/ml) compared to poor responders (PSA ≥0.1 ng/ml) was assessed.^{12–14}

Image acquisition

All baseline MRI examinations were performed on a GE Signa HDx 3.0T magnet (GE Healthcare, Waukesha, WI, USA) using a combination of an eight-channel abdominal array and endorectal coil (Medrad, Pittsburgh, PA, USA). The multiparametric protocol³⁸ included T1- and T2-weighted imaging, diffusion-weighted imaging (DWI; b = 0 and 1,400 s/mm²) and dynamic contrast-enhanced (DCE) MRI. DCE-MRI utilised a 3D spoiled gradient recalled echo (SPGR) sequence using 3.6 ms repetition time (TR), 1.3 ms echo time (TE), $\alpha = 15^\circ$, $26 \times 26 \text{ cm}^2$ field of view (FOV), with full gland coverage and an interpolated voxel size of $1 \times 1 \times 6 \text{ mm}^3$, with frames acquired at approximately 5-second intervals. Gadopentetate dimeglumine (Magnevist, Berlex Laboratories, Wayne, NJ, USA) was injected intravenously using a syringe pump (0.15 mmol/kg; rate 3 ml/s). ADC maps were calculated using DWI by performing a mono-exponential fit to the data acquired with b-values 0–1,400 s/mm², in conformity with PI-RADS version 2 recommendations.³⁹ After obtaining the standard of care baseline study with endorectal coil, all subsequent prostate MRI examinations were obtained using the above protocol but were modified for pelvic coil only (non-endorectal study).

Image analysis

All MRI studies were de-identified, and tumours were identified and contoured by a freehand region of interest (ROI) tool using 3D Slicer (www.slicer.org), as previously described.⁴⁰ Images were evaluated prospectively by a radiologist (>15 years of experience in prostate mpMRI), who was aware of the study inclusion criteria and had the patient's clinical and pathological data available at baseline. Index tumours were selected using standard PI-RADS v. 2³⁹ criteria for clinically significant prostate cancer, i.e., those who received a category ≥4. In case of multiple lesions with PI-RADS score ≥4, the lesion with the lowest ADC value was selected.

Mean tumour ADC (tADC) and tumour volume (tVOL) were obtained for all TPs. A non-tumour “normal” ROI, based upon appearance on all sequences, was also contoured on ADC maps for each patient to extract normal quantitative ADC metrics (nADC), at all TP in both the peripheral (PZ) and transition (TZ) zones, as previously described.¹⁷ Mean ADC was calculated for each ROI. Whole prostate gland volume was also measured at all TPs, volumetrically contoured on axial T2-weighted images.

Statistical analysis

The baseline and post-treatment values of tADC and nADC were compared on a per patient basis in a sequential fashion (i.e., baseline tADC versus TP1, TP1 versus TP2 and TP2 versus TP3) using two-tailed paired Student's *t*-test. A similar approach was taken for comparison of tVOL, prostate gland volume, and PSA.

Absolute and relative changes (percentage) in ADC metrics, tVOL and prostate gland volume were correlated with serum PSA using Pearson's correlation coefficient. A two-

sided p -value of <0.05 was considered to indicate statistical significance. SPSS v. 24 was used for all statistical analyses.

Results

Patient and clinical characteristics

Twelve patients with biopsy-proven clinically significant PCa consented to the study and completed all four mpMRIs. The mean patient age at the time of diagnosis was 70.1 (range 56–79) years of age. Mean serum PSA at baseline was 24.7 ng/ml (SD $\pm$ 32; range 4.4–119.7 ng/ml), with a mean PSA density of 0.4 ng/ml² (SD $\pm$ 0.04, range 0.11–1.44 ng/ml²). Fifteen clinically significant lesions were identified using PI-RADS v. 2 criteria. Among those, 12 index lesions were selected, one per patient. Six were identified in the PZ and six in the TZ. All mpMRIs from each TP were of sufficient quality for interpretation, except for ADC maps for patient 10 in TP1 due to imaging artefacts. As such, for comparative analysis at TP1, ADC values for patient 10 was treated as missing data. Patients were treated for a mean of 14.8 months with nADT (range 6e26 months). A summary of the baseline demographic data is provided in Table 1.

The mean interval between ADT initiation and TP1 MRI was 69.7 days (median 64.5 days) and between initiation of EBRT and TP2 MRI was 41.7 days (median 42 days). Clinical outcome was followed for 5 years after treatment initiation. One patient had disease recurrence, characterised by an increased PSA level of 6.87 ng/ml, which occurred 18 months after completing 6 months of nADT therapy.

PSA dynamics

Serum PSA levels were available for all patients at all TPs except for patient 12, whose PSA level at TP1 was not obtained to coincide with the MRI and was considered missing data in the paired comparisons involving TP1.

Overall, there was a significant decrease in serum PSA levels at all TPs compared in a sequential fashion for all patients (Fig 2). Paired t -test demonstrated a decreased PSA, which at baseline was 26 ± 33.2 and 1.2 ± 1.5 , 0.3 ± 0.3 , and 0.1 ± 0.09 ng/ml at TPs 1, 2, and 3, respectively; $p < 0.05$ for all comparisons with the preceding TP. Similarly, PSA density significantly decreased from baseline compared to all subsequent TPs. Mean PSA density was 0.4 ± 0.04 ng/ml² at baseline, 0.03 ± 0.03 ng/ml² at TP1, 0.01 ± 0.01 ng/ml² at TP2, and 0.004 ± 0.005 ng/ml² at TP3; $p < 0.05$ for all comparisons with the preceding TP.

All patients reached their lowest PSA value post-treatment at TP3 (post-treatment PSA), which ranged from 0.02 to 0.32 ng/ml. Using a cut-off of 0.1 ng/ml, seven patients had a post-treatment PSA < 0.1 ng/ml and five patients ≥ 0.1 ng/ml at TP3.

ADC metrics

ADC signal in patient 10 was excluded from the analysis at TP1 due to image artefacts.

Comparison of tADC with its corresponding PZ or TZ nlADC demonstrated the tADC to be significantly lower than nlADC at all TPs, as outlined in Table 2. tADC compared with the corresponding nlADC ($p < 0.01$ for all comparisons).

Patient 10 was excluded from the sequential analysis due to missing data at TP1. The reported means used for comparison at baseline and TP2 was based on the remaining 11 patients. There was a significant increase in tADC from baseline to TP1 (917.8 ± 107.7 versus $1033.8 \pm 139.3 \times 10^{-6} \text{ mm}^2/\text{s}$, $p = 0.01$); however, there was no subsequent sequential change in tADC, as outlined in Table 3. The relative increase (%) in tADC from BL to post-treatment TP3 had a significant correlation with absolute tADC value at baseline ($r = -0.670$; $p = 0.01$), with those lesions with a lower tADC value at BL demonstrating greater tADC relative increase from BL to TP3.

A non-significant downtrend in nlADC was observed during therapy at all TPs for both the TZ and PZ. Changes in tADC and nlADC in the PZ and TZ are summarised in Fig 3.

Tumour volume and prostate gland volume

As outlined in Fig 4, mean tVOL decreased from BL to TP1 (9.35 ± 10.2 versus $2.90 \pm 3.5 \text{ cm}^3$; $p = 0.02$) and from TP1 to TP2 (2.90 ± 3.5 versus $0.75 \pm 0.5 \text{ cm}^3$; $p = 0.04$). No subsequent change was seen at TP3. Prostate gland volume also decreased significantly from BL to TP1 (58.4 ± 32 versus $43.6 \pm 28.9 \text{ cm}^3$; $p < 0.01$), and from TP2 to TP3, (43.2 ± 29.4 versus $39.1 \pm 25.4 \text{ cm}^3$; $p = 0.01$). The relative decrease (%) in tumour volume from BL to TP1 correlated with post-treatment PSA ($r = 0.635$; $p = 0.03$).

Correlation of post-treatment PSA with changes in tumour ADC

Seven patients had a post-treatment PSA $< 0.1 \text{ ng/ml}$, and five patients had a post-treatment PSA $\geq 0.1 \text{ ng/ml}$. Post-treatment PSA correlated with both the absolute and relative (%) change in tADC from BL to TP1 ($r = -0.674$ and $r = 0.666$ respectively; $p = 0.02$), as outlined in Fig 5. Those with lower post-treatment PSA had a greater change in tADC from BL to TP1 compared to those with a higher post-treatment PSA ($169.2 \pm 122.4 \times 10^{-6}$ versus $22.9 \pm 75.5 \times 10^{-6} \text{ mm}^2/\text{s}$, $p = 0.03$). Electronic Supplementary Material Figs 1 and 2 show examples of good and poor therapy responders.

Correlation of changes in normal prostate tissue with PSA

There was no significant correlation between prostate gland volume changes and either post-treatment PSA or post-treatment PSA density at TP3. Similarly, no absolute value of nlADC or its dynamic changes during the entire study correlated with post-treatment PSA.

Discussion

The present prospective study demonstrated that early dynamic changes in tADC correlate with post-treatment serum PSA, obtained 6-months after combined neoadjuvant therapy and EBRT. Moreover, these early dynamic ADC changes differ significantly among good versus poor therapy responders, as defined by post-treatment PSA criteria. These criteria

for response are known to predict clinical outcomes such as risk of recurrence and death.^{12–14} The importance of these findings is underscored by the fact that as early changes in tADC, particularly changes detected prior to initiation of EBRT, are able to predict treatment response, treatment intensification and length could be adjusted during neoadjuvant therapy. This may have important ramifications for appropriately selecting patients for treatment intensification (e.g., radiation dose escalation, brachytherapy boost, and intensification of anti-androgen therapy) if initial tADC change is suboptimal.

Post-treatment PSA did not correlate with the absolute value or dynamic changes in the nlADC nor with whole-gland volume reduction during therapy. One could not simply attribute the PSA reduction to changes in normal prostate tissue. To the authors' knowledge, this study is novel in that it assessed prospective and serial dynamic ADC changes in a population that have undergone combined nADT prior to EBRT and explore the correlation of ADC with post-treatment biochemical response.

In the present cohort, baseline tADC was lower than nlADC, with values similar to those previously reported in the literature.^{31,32} After initiation of neoadjuvant therapy, an overall increase in tADC and a decrease in nlADC was observed, similar to previous studies.^{32,35–37,41,42} There was an early increase in tADC in response to nADT, but no subsequent increase during the course nor after EBRT delivery. The differences in tADC during EBRT may be attenuated after hormonal therapy, as previously suggested by McPartlin *et al.*³³ Even though it has been suggested that good responders to EBRT have a more marked change in tADC than non-responders,^{37,41} and that the changes in tADC between baseline and 6 months after EBRT could predict PSA decrease at least 12 months in advance,³⁷ this is not the case in a PCa population undergoing combined nADT and EBRT.

Others have investigated temporal changes of ADC in tumour foci and normal appearing tissue during and post-radiation therapy. These studies overall show an increase in tADC both in the short- and long-term follow-up^{31,32,34,37,41–43}; however, hormone therapy was not administered prior to EBRT^{34,37,43} or was delivered to only a subpopulation.⁴²

Hotker *et al.*³⁵ and Kim *et al.*³⁶ found a significant increase in tADC postnADT in their retrospective studies, similarly to the present study. In a similar prospective study, Barret *et al.*²⁴ found median tADC did not significantly change after nADT, although this study looked at global tADC measurement rather than that of the index lesion.

On a cellular level, nADT has been shown to impair prostatic blood flow and oxygenation, causing nuclear shrinkage, cell vacuolisation, and apoptosis and inducing acinar atrophy, basal cell hyperplasia and fibrosis in the glandular stromal tissue⁴⁴; with a somewhat different effect on tumour cells from normal tissue.⁴⁵ These cellular mechanism changes may be represented by the observed early significant tADC increase in the tumour foci after initiation on nADT. Moreover, tumour volume decreased significantly during early nADT and subsequently after EBRT delivery, possibly reflecting the sensitivity of PCa to androgen therapy in the present population.

The present study also determined how dynamic changes during early nADT correlated with clinical response, as defined by post-treatment PSA. Prior large clinical studies have shown

that a PSA <0.1 ng/ml after combined neoadjuvant therapy is associated with improved clinical outcomes measured such as cancer-specific survival, biochemical progression-free survival, and overall survival.^{12–14} Using this cut-off, those with a lower post-treatment PSA had a significantly higher increase in the absolute and relative (%) tADC in TP1 compared to baseline, possibly reflecting early treatment response at the cellular level.

Even though absolute ADC values have been correlated with tumour aggressiveness and is well known to reflect tumour cellularity at a histological level, its value added little as a predictor in the present study. These different outcomes suggest that absolute ADC value may have a limited role in assessing tumour response, whereas its dynamic changes during therapy may add value as suggested by some authors.^{35–37}

Finally, concern has been brought in the literature regarding the ADC variation across multiple vendors and lack of standardisation. In the present study, all the baseline and follow-up scans were performed at pre-established intervals in the same scanner with similar imaging technique, accounting for this possible limitation. Moreover, a recent study from Moller *et al.*⁴⁶ suggested that the variation of ADC across multiple vendors is limited and suggested of non-clinical significance.

The present study had several limitations. First, the study population was small, composed of 12 patients with one index lesion per patient. Although intermediate to a high-risk population was studied, there was heterogeneity in the Gleason score, PSA, and tumour volume at Baseline. Second, due to patient discomfort, the use of an endorectal coil was a concern for the Institutional Review Board, and therefore, was not used in the follow-up studies. It could potentially have improved the signal-to-noise ratio and provide better accuracy when delineating tumour margins, especially after prostate shrinkage after androgen-deprivation therapy and EBRT. Finally, imaging findings were analysed by just one experienced reader, which limits the evaluation of inter-reader agreement. Although the radiologist was blinded to the outcomes, it was known that all cases harboured clinically significant disease from the outset and lesions were evident at baseline.

In conclusion, this pilot prospective study demonstrates that early dynamic ADC changes within index prostate tumour foci on MRI may serve as a biomarker of clinical response to combined nADT and EBRT. Further studies in a larger cohort are warranted to provide validation and expansion of these findings.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgements

The authors thank Anthony D'Amico, MD, and Clair Beard, MD. This work was supported by the National Institutes of Health grants nos. U01CA151261 and P41EB028741 (authors A.F., M.V. and F.M.F.). F.M.F. was also supported by T32EB025823.

References

1. Bray F, Ferlay J, Soerjomataram I, et al. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 2018;68:394–424. [PubMed: 30207593]
2. Society AC. Key statistics for prostate cancer. American Cancer Society Website; 2018 <https://www.cancer.org/cancer/prostate-cancer/about/key-statistics.html>. [Accessed 30 July 2023].
3. D'Amico AV. Biochemical outcome after radical prostatectomy, external beam radiation therapy, or interstitial radiation therapy for clinically localized prostate cancer. *JAMA* 1998;280:969–74. 10.1001/jama.280.11.969. [PubMed: 9749478]
4. Mohler JL, Kantoff PW, Armstrong AJ, et al. Prostate cancer, version 2.2014. *J Natl Compr Canc Netw* 2014;12:686–718. [PubMed: 24812137]
5. Bolla M, Van Tienhoven G, Warde P, et al. External irradiation with or without long-term androgen suppression for prostate cancer with high metastatic risk: 10-year results of an EORTC randomised study. *Lancet Oncol* 2010;11:1066–73. [PubMed: 20933466]
6. Pilepich MV, Caplan R, Byhardt RW, et al. Phase III trial of androgen suppression using goserelin in unfavorable-prognosis carcinoma of the prostate treated with definitive radiotherapy: report of Radiation Therapy Oncology Group Protocol 85–31. *J Clin Oncol* 1997;15:1013–21. 10.1200/jco.1997.15.3.1013. [PubMed: 9060541]
7. D'Amico AV, Chen M-H, Renshaw AA, et al. Androgen suppression and radiation versus radiation alone for prostate cancer: a randomized trial. *JAMA* 2008;299:289–95. [PubMed: 18212313]
8. Warde P, Mason M, Ding K, et al. Combined androgen deprivation therapy and radiation therapy for locally advanced prostate cancer: a randomised, phase 3 trial. *Lancet* 2011;378:2104–11. [PubMed: 22056152]
9. Widmark A, Klepp O, Solberg A, et al. Endocrine treatment, with or without radiotherapy, in locally advanced prostate cancer (SPCG-7/SFUO-3): an open randomised phase III trial. *Lancet* 2009;373:301–8. [PubMed: 19091394]
10. Brenner H, Arndt V. Long-term survival rates of patients with prostate cancer in the prostate-specific antigen screening era: population-based estimates for the year 2000 by period analysis. *J Clin Oncol* 2005;23:441–7. [PubMed: 15572727]
11. D'Amico AV, Moul J, Carroll PR, et al. Cancer-specific mortality after surgery or radiation for patients with clinically localized prostate cancer managed during the prostate-specific antigen era. *J Clin Oncol* 2003;21:2163–72. [PubMed: 12775742]
12. Bryant AK, D'Amico AV, Nguyen PL, et al. Three-month posttreatment prostate-specific antigen level as a biomarker of treatment response in patients with intermediate-risk or high-risk prostate cancer treated with androgen deprivation therapy and radiotherapy. *Cancer* 2018;124:2939–47. [PubMed: 29727915]
13. Narang AK, Trieu J, Radwan N, et al. End-of-radiation PSA as a novel prognostic factor in patients undergoing definitive radiation and androgen deprivation therapy for prostate cancer. *Prostate Cancer Prostatic Dis* 2017;20:203–9. [PubMed: 28094250]
14. Naik M, Reddy CA, Stephans KL, et al. Posttreatment prostate-specific antigen 6 months after radiation with androgen deprivation therapy predicts for distant metastasis-free survival and prostate cancer-specific mortality. *Int J Radiat Oncol Biol Phys* 2016;96:617–23. [PubMed: 27681757]
15. Adhyam M, Gupta AK. A review on the clinical utility of PSA in cancer prostate. *Indian J Surg Oncol* 2012;3:120–9. [PubMed: 23730101]
16. Zilli T, Dal Pra A, Kountouri M, et al. Prognostic value of biochemical response to neoadjuvant androgen deprivation before external beam radiotherapy for prostate cancer: a systematic review of the literature. *Cancer Treat Rev* 2016;46:35–41. [PubMed: 27082769]
17. Hassanzadeh E, Alessandrino F, Olubiyi OI, Glazer DI, Mulkern RV, Fedorov A, et al. Comparison of quantitative apparent diffusion coefficient parameters with prostate imaging reporting and data system V2 assessment for detection of clinically significant peripheral zone prostate cancer. *Abdom Radiol (NY)* 2018;43:1237–44. [PubMed: 28840280]

18. Alessandrino F, Taghipour M, Hassanzadeh E, et al. Predictive role of PI-RADSV2 and ADC parameters in differentiating Gleason pattern 3 + 4 and 4 + 3 prostate cancer. *Abdom Radiol (NY)* 2019;44:279–85. [PubMed: 30066169]
19. Beksac AT, Kumarasamy S, Falagario U, et al. Multiparametric magnetic resonance imaging features identify aggressive prostate cancer at the phenotypic and transcriptomic level. *J Urol* 2018;200:1241–9. [PubMed: 30563651]
20. Falagario UG, Beksac AT, Martini A, et al. Defining prostate cancer at favorable intermediate risk: the potential utility of magnetic resonance imaging and genomic tests. *J Urol* 2019;202:102–7. [PubMed: 30730408]
21. Moore CM, Giganti F, Albertsen P, et al. Reporting magnetic resonance imaging in men on active surveillance for prostate cancer: the PRECISE recommendationsda report of a European School of Oncology Task Force. *Eur Urol* 2017;71:648–55. [PubMed: 27349615]
22. Fennessy FM, Fedorov A, Vangel MG, et al. Multiparametric MRI as a biomarker of response to neoadjuvant therapy for localized prostate cancer—a pilot study. *Acad Radiol* 2019, 10.1016/j.acra.2019.10.017.
23. Fennessy FM, McKay RR, Beard CJ, et al. Dynamic contrast-enhanced magnetic resonance imaging in prostate cancer clinical trials: potential roles and possible pitfalls. *Transl Oncol* 2014;7:120–9. [PubMed: 24772215]
24. Barrett T, Gill AB, Kataoka MY, et al. DCE and DW MRI in monitoring response to androgen deprivation therapy in patients with prostate cancer: a feasibility study. *Magn Reson Med* 2012;67:778–85. [PubMed: 22135228]
25. Hambrock T, Somford DM, Huisman HJ, et al. Relationship between apparent diffusion coefficients at 3.0-T MR imaging and Gleason grade in peripheral zone prostate cancer. *Radiology* 2011;259:453–61. [PubMed: 21502392]
26. Turkbey B, Shah VP, Pang Y, et al. Is apparent diffusion coefficient associated with clinical risk scores for prostate cancers that are visible on 3-T MR images? *Radiology* 2011;258:488–95. [PubMed: 21177390]
27. Tan CH, Wei W, Johnson V, et al. Diffusion-weighted MRI in the detection of prostate cancer: meta-analysis. *AJR Am J Roentgenol* 2012;199:822–9. [PubMed: 22997374]
28. Verma S, Rajesh A, Morales H, et al. Assessment of aggressiveness of prostate cancer: correlation of apparent diffusion coefficient with histologic grade after radical prostatectomy. *AJR Am J Roentgenol* 2011;196:374–81. [PubMed: 21257890]
29. Itou Y, Nakanishi K, Narumi Y, et al. Clinical utility of apparent diffusion coefficient (ADC) values in patients with prostate cancer: can ADC values contribute to assess the aggressiveness of prostate cancer? *J Magn Reson Imaging* 2011;33:167–72. [PubMed: 21182135]
30. Vargas HA, Akin O, Franiel T, et al. Diffusion-weighted endorectal MR imaging at 3 T for prostate cancer: tumour detection and assessment of aggressiveness. *Radiology* 2011;259:775–84. [PubMed: 21436085]
31. Foltz WD, Wu A, Chung P, et al. Changes in apparent diffusion coefficient and T2 relaxation during radiotherapy for prostate cancer. *J Magn Reson Imaging* 2013;37:909–16. [PubMed: 23097411]
32. Park SY, Kim CK, Park BK, et al. Early changes in apparent diffusion coefficient from diffusion-weighted MR imaging during radiotherapy for prostate cancer. *Int J Radiat Oncol Biol Phys* 2012;83:749–55. [PubMed: 22154286]
33. McPartlin A, Kershaw L, McWilliam A, et al. Changes in prostate apparent diffusion coefficient values during radiotherapy after neoadjuvant hormones. *Ther Adv Urol* 2018;10:359–64. [PubMed: 30574195]
34. Decker G, Muertz P, Gieseke J, et al. Intensity-modulated radiotherapy of the prostate: dynamic ADC monitoring by DWI at 3.0 T. *Radiother Oncol* 2014;113:115–20. [PubMed: 25304719]
35. Hötker AM, Mazaheri Y, Zheng J, et al. Prostate cancer: assessing the effects of androgen-deprivation therapy using quantitative diffusion-weighted and dynamic contrast-enhanced MRI. *Eur Radiol* 2015;25:2665–72. [PubMed: 25820537]

36. Kim AY, Kim CK, Park SY, et al. Diffusion-weighted imaging to evaluate for changes from androgen deprivation therapy in prostate cancer. *AJR Am J Roentgenol* 2014;203:W645–50. [PubMed: 25415730]
37. Pasquier D, Hadj Henni A, Escande A, et al. Diffusion weighted MRI as an early predictor of tumour response to hypofractionated stereotactic boost for prostate cancer. *Sci Rep* 2018;8:10407. [PubMed: 29991748]
38. Hegde JV, Mulkern RV, Panych LP, et al. Multiparametric MRI of prostate cancer: an update on state-of-the-art techniques and their performance in detecting and localizing prostate cancer. *J Magn Reson Imaging* 2013;37:1035–54. [PubMed: 23606141]
39. Weinreb JC, Barentsz JO, Choyke PL, et al. PI-RADS prostate imaging - reporting and data system: 2015, version 2. *Eur Urol* 2016;69:16–40. [PubMed: 26427566]
40. Fedorov A, Beichel R, Kalpathy-Cramer J, Finet J, Fillion-Robin J-C, Pujol S, et al. 3D Slicer as an image computing platform for the Quantitative Imaging Network. *Magn Reson Imaging* 2012;30:1323–41. [PubMed: 22770690]
41. Liu L, Wu N, Ouyang H, et al. Diffusion-weighted MRI in early assessment of tumour response to radiotherapy in high-risk prostate cancer. *Br J Radiol* 2014;87:20140359. [PubMed: 25162831]
42. Song I, Kim CK, Park BK, et al. Assessment of response to radiotherapy for prostate cancer: value of diffusion-weighted MRI at 3 T. *AJR Am J Roentgenol* 2010;194:W477–82. [PubMed: 20489065]
43. Takayama Y, Kishimoto R, Hanaoka S, et al. ADC value and diffusion tensor imaging of prostate cancer: changes in carbon-ion radiotherapy. *J Magn Reson Imaging* 2008;27:1331–5. [PubMed: 18504751]
44. Roznovanu SL, Rădulescu D, Novac C, et al. The morphologic changes induced by hormone and radiation therapy on prostate carcinoma. *Rev Med Chir Soc Med Nat Iasi* 2005;109:337–42. [PubMed: 16607796]
45. Reuter VE. Pathological changes in benign and malignant prostatic tissue following androgen deprivation therapy. *Urology* 1997;49:16–22. [PubMed: 9123731]
46. Møller JM, Boesen L, Hansen AE, et al. Quantification of cross-vendor variation in ADC measurements in vendor-specific prostate MRI-protocols. *Eur J Radiol* 2023 Jun 21:110942.

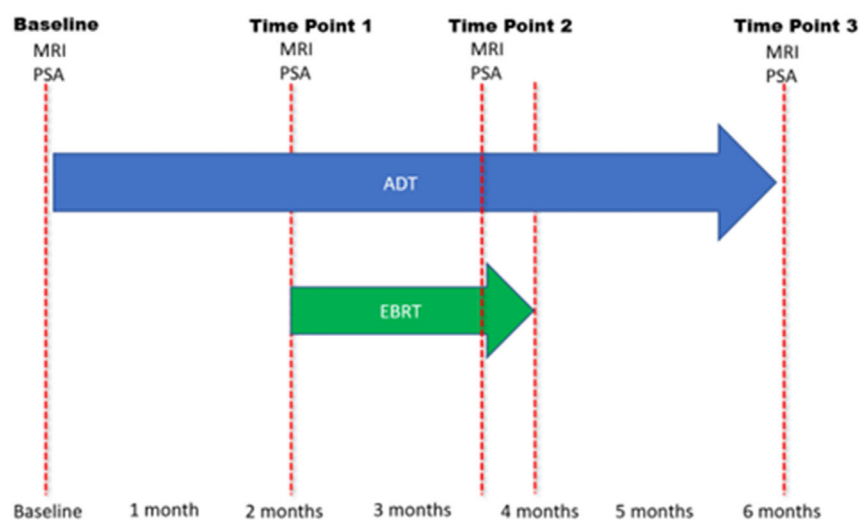


Figure 1. Timeline of the study. MRI and PSA were obtained at Baseline, 8 weeks after nADT initiation, 6 weeks into EBRT and 6 months after initiation of treatment.

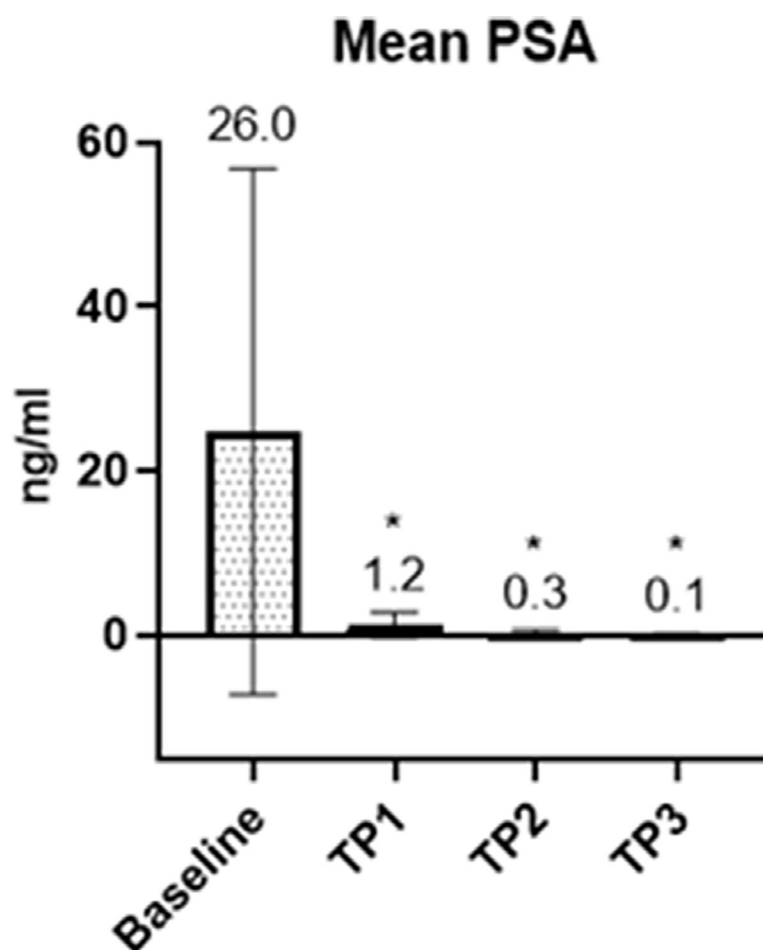


Figure 2. Serum PSA values at baseline and all subsequent TPs (TP1–3) during the study. The asterisk denotes a significant change ($p < 0.05$) in mean serum PSA compared with the preceding TP.

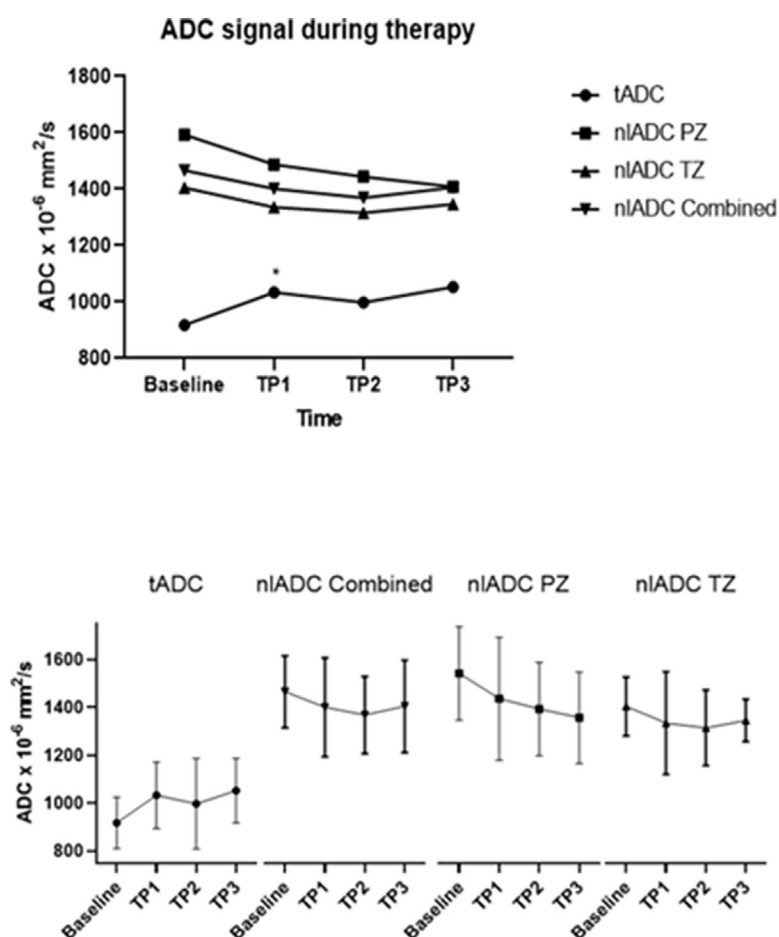


Figure 3.

Changes in ADC for normal tissue and tumour at the different TPs (TP1–3). The asterisk denotes a significant change ($p < 0.05$) compared with the antecessor TP.

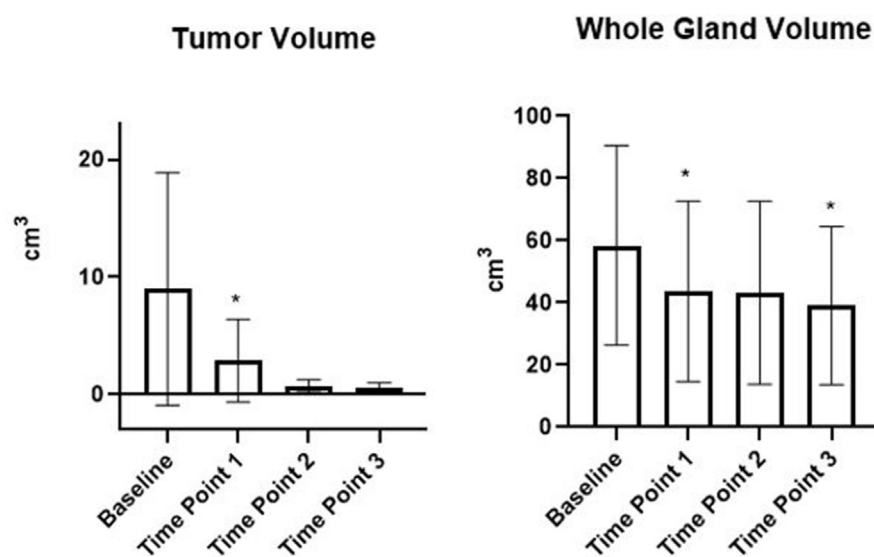


Figure 4. Tumour and whole gland volume in the different TPs during therapy. The asterisk denotes a significant change ($p < 0.05$) compared with the antecessor TP.

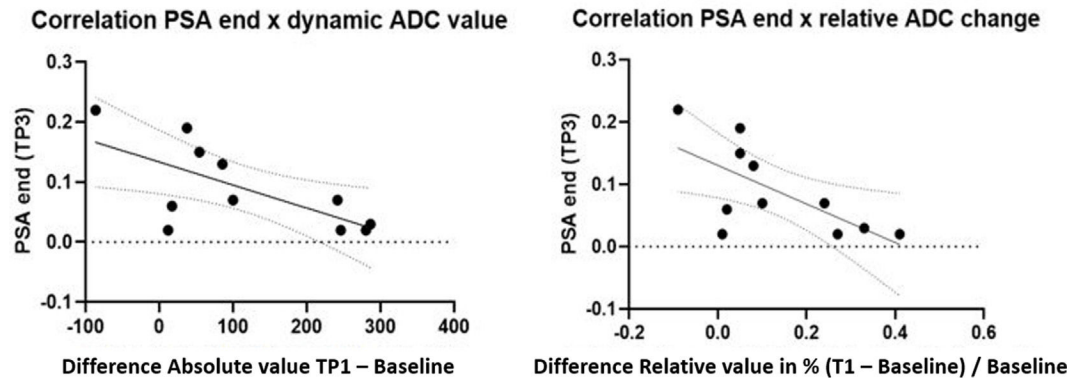


Figure 5.

Correlation of the early tumour ADC at TP1 with PSA at the end of therapy. (a) tADC absolute values changes and (b) relative changes as a percentage. Overall, tumours with greater absolute increase in absolute or relative ADC in TP1 presented a lower PSA at the end of the study.

Table 1

Patient clinical and pathology characteristics at baseline.

Patient	Age (years)	Tumour	Location	Gleason score	Tumour volume (cm ³)	PSA	Clinical stage	PSA density (ng/ml ²)
01	71	PZ	Right	4 + 5	5.4	11.3	T3b	0.24
02	67	PZ	Bilateral	4 + 5	24.8	4.4	T2c	0.11
03	71	TZ	Left	4 + 5	0.4	15	T2b	0.54
04	77	TZ	Left	4 + 3	0.4	6.1	T2a	0.14
05	62	TZ	Left	3 + 4	3.8	11.6	T1c	0.11
06	78	PZ	Left	3 + 4	0.9	7	T2a	0.14
		TZ	Right	3 + 3	0.7	-		
07	70	PZ	Left	4 + 5	3.9	119.7	T2c	1.44
		PZ	Right	4 + 5	1.7	-		
		TZ	Left	4 + 4	13.9	-		
08	73	TZ	Right	3 + 3	6.1	42	T1c	0.83
09	71	PZ	Right	4 + 3	3.4	12.5	T3a	0.3
10	67	PZ	Right	5 + 4	4.8	33.2	T3a	0.92
11	79	PZ	Left	4 + 4	30.7	24	T3b	0.18
12	56	PZ	Left	4 + 4	13	10.1	T3a	0.24

Tumour location PZ stands for “peripheral zone” and TZ for “transition zone”.
PSA, prostate-specific antigen.

Table 2

Comparison of ADC signal values for tumour foci and corresponding normal prostate tissue at baseline and all subsequent time points (TP1—3) during the study.

	nlADC ($\times 10^{-6}$ mm²/s $\pm$SD)	tADC ($\times 10^{-6}$ mm²/s $\pm$SD)	p-Value
Baseline	896.8 $\pm$ 125.8	1444.5 $\pm$ 162.4	<0.01
TP1	1033.8 $\pm$ 139.3	1401 $\pm$ 206.5	<0.01
TP2	997.5 $\pm$ 189.6	1368.7 $\pm$ 160.8	<0.01
TP3	1053.1 $\pm$ 134.3	1405.5 $\pm$ 192.4	<0.01

Comparison of cohort mean ADC signal values for tumour foci and normal prostate tissue at baseline and all subsequent time points (TP1—3) during the study.

Table 3

	Baseline	TP1	TP2	TP3
tADC	917.8 ± 107.7 ^a	1033.8 ± 139.3 ^a	1012.3 ± 191.5	1053.1 ± 134.3
nlADC PZ	1593.2 ± 195.1	1487.1 ± 256.1	1443.5 ± 196.8	1408.1 ± 191.1
nlADC TZ	1404.4 ± 122.4	1335.1 ± 214.2	1323.7 ± 162.2	1346.4 ± 87.8
nlADC combined	1444.5 ± 162.4	1401.0 ± 206.5	1368.7 ± 160.8	1405.5 ± 192.4

Mean values (±SD) are given in 10^{−6} mm²/s at baseline and at different time points (TP1—3) in tumour (tADC) and benign tissue (nlADC).

^aDenotes a significant change (*p*<0.05).